ORIGINAL PROJECT

Finding Minimum Travel Times from a Selected Bus Stop Using Dijkstra's Algorithm

Problem Statement

A bus network contains stops connected by routes with different travel times. Finding the minimum travel time from one selected stop to all other reachable stops can be difficult when many routes exist. A systematic graph-based method is needed to calculate the shortest travel time.

Solution

Represent each bus stop as a vertex and each direct connection as a weighted edge, with travel time as its weight. Set the selected source stop's distance to zero and all other distances to infinity. Repeatedly choose the unvisited stop with the smallest known distance and update its neighbours when a shorter route is found. Finally, display the minimum travel time from the source to every reachable stop, and report when a stop cannot be reached.